四大发明

造纸术·印刷术·指南针·火药

The Four Great Inventions

Paper · Printing · The Compass · Gunpowder

Text and translation by Claude (Anthropic)

Footnotes mark names, festival terms and words above HSK 4 (HSK 3.0 lists: github.com/ivankra/hsk30, MIT).

1 在中国，几乎每个学生都听说过“四大发明”[1]：造纸术[2]、印刷术[3]、指南针[4]和火药[5]。这四种技术都是很早以前在中国出现的，后来又慢慢传到了世界上别的地方。它们改变了人们写字、读书、旅行和打仗[6]的方法，也改变了历史。下面我们就一个一个地来看看它们的故事。

造纸术

2 在纸出现以前，中国人把字写在什么上面呢？最常用的是竹简[7]，就是把竹子[8]切成长长的、窄窄[9]的片，再用绳子[10]连在一起。竹简很重，一本书常常要用车来拉。有钱人也会在丝绸[11]上写字，丝绸又轻又软[12]，可是太贵了，一般人用不起。

1 四大发明 *sì dà fāmíng* Four Great Inventions
2 造纸术 *zàozhǐshù* papermaking
3 印刷术 *yìnshuāshù* printing
4 指南针 *zhǐnánzhēn* compass
5 火药 *huǒyào* gunpowder
6 打仗 *dǎzhàng* to fight a war
7 竹简 *zhújiǎn* bamboo slips (for writing)
8 竹子 *zhúzi* bamboo
9 窄窄 *zhǎizhǎi* narrow
10 绳子 *shéngzi* string, cord
11 丝绸 *sīchóu* silk
12 软 *ruǎn* soft

Almost every student in China has heard of the "Four Great Inventions": 1
papermaking, printing, the compass and gunpowder. All four technologies first appeared in China long ago and later slowly spread to other parts of the world. They changed the way people wrote, read, travelled and fought wars, and they changed history too. Let us look at their stories one by one.

Papermaking

Before paper appeared, what did Chinese people write on? The most com- 2
mon material was bamboo slips: bamboo cut into long, narrow strips and then tied together with cord. Bamboo slips were heavy; a single book often had to be carried on a cart. The rich also wrote on silk, which is light and soft, but it was far too expensive for ordinary people.

3 考古学家发现，早在两千多年前的西汉[1]，中国就已经有了一种很粗的纸，不过当时的纸主要用来[2]包东西，不太适合写字。到了东汉[3]，一位叫蔡伦[4]的官员[5]改进了造纸的方法。据说在公元 105 年左右，他把树皮、麻[6]、旧布和破渔网[7]放在水里泡[8]烂[9]，再把它们打成纸浆[10]。然后，人们用一张细细的竹帘[11]把纸浆从水里捞[12]起来，等水流走，纸浆就变成了一层薄薄的纸。这种纸又轻又便宜，写字也很方便，所以很快就在全国流行起来。

4 很长一段时间里，造纸的方法只有中国人知道。后来，这种技术先传到了朝鲜半岛[13]和日本，又沿着[14]丝绸之路[15]向西传播。

1 西汉 *Xī Hàn* Western Han dynasty
2 用来 *yònglái* to be used for
3 东汉 *Dōng Hàn* Eastern Han dynasty
4 蔡伦 *Cài Lún* Cai Lun, Eastern Han court official
5 官员 *guānyuán* official
6 麻 *má* hemp
7 渔网 *yúwǎng* fishing net
8 泡 *pào* to soak
9 烂 *làn* to go soft
10 纸浆 *zhǐjiāng* paper pulp
11 竹帘 *zhúlián* bamboo screen
12 捞 *lāo* to scoop up
13 朝鲜半岛 *Cháoxiǎn Bàndǎo* Korean peninsula
14 沿着 *yánzhe* along
15 丝绸之路 *Sīchóu zhī Lù* the Silk Road

Archaeologists have found that as early as the Western Han, more than two thousand years ago, China already had a kind of coarse paper; but it was used mainly for wrapping things and was not very good for writing. In the Eastern Han, an official named Cai Lun improved the way paper was made. It is said that around AD 105 he soaked tree bark, hemp, old cloth and worn-out fishing nets in water until they went soft, and then beat them into pulp. Then a fine bamboo screen was used to lift the pulp out of the water; once the water had drained away, the pulp had become a thin sheet of paper. This paper was light and cheap, and easy to write on, so it soon became popular all over the country. 3

For a long time, only the Chinese knew how to make paper. Later the technique spread first to the Korean peninsula and Japan, and then westward along the Silk Road. 4

4 据说在公元 751 年的一场战争以后，一些会造纸的中国工人被带到了中亚[1]的撒马尔罕[2]，造纸术[3]从那里传到了阿拉伯[4]世界，几百年以后又到了欧洲[5]。有了便宜的纸，书变得越来越多，知识也更容易传给更多的人。

印刷术[6]

5 有了纸，还需要一种快速做书的方法。以前，每一本书都得用手一个字一个字地抄，抄一本书要花很多天，而且很容易抄错。大概在唐代[7]，人们发明了雕版印刷[8]。他们先把一页书的内容反着刻在一块木板[9]上，然后在木板上刷上墨[10]，再把纸放上去轻轻地压，一页书就印[11]好了。

1 中亚 *Zhōng Yà* Central Asia
2 撒马尔罕 *Sāmǎ'ěrhǎn* Samarkand
3 造纸术 *zàozhǐshù* papermaking
4 阿拉伯 *Ālābó* Arab, Arabia
5 欧洲 *Ōuzhōu* Europe
6 印刷术 *yìnshuāshù* printing
7 唐代 *Tángdài* Tang dynasty (618–907)
8 雕版印刷 *diāobǎn yìnshuā* woodblock printing
9 木板 *mùbǎn* wooden board
10 墨 *mò* ink
11 印 *yìn* to print

It is said that after a battle in 751, some Chinese papermakers were taken to 4
Samarkand in Central Asia; from there papermaking reached the Arab world, and several hundred years later Europe. With cheap paper, books became more and more common, and knowledge could be passed on to many more people.

Printing

Once there was paper, a quick way of making books was still needed. Before 5
that, every book had to be copied by hand, character by character; copying one book took many days, and mistakes crept in easily. Probably during the Tang dynasty, people invented woodblock printing. First they carved the text of a whole page, in reverse, into a wooden block; then they brushed ink onto the block, laid paper on top and pressed it gently, and the page was printed.

5 同[1]一块木板[2]可以印[3]很多次，所以一本书可以印出成百上千[4]份。

6 现在能看到的、最早的、有明确日期的印刷[5]书，是一本唐代[6]的《金刚经[7]》，上面的日期相当于[8]公元 868 年。这本书是在敦煌[9]的一个山洞[10]里发现的，今天保存在英国伦敦[11]的大英图书馆[12]里。不过，雕版印刷[13]也有缺点：每印一本新书，都要刻一套新的木板，又费时间又费力气。

7 到了北宋[14]，一个叫毕昇[15]的普通工匠[16]想出了一个新办法。他用黏土[17]做成很多小方块，每个方块上刻一个字，然后用火烧硬[18]，这就是活字[19]。

1 同 *tóng* same
2 木板 *mùbǎn* wooden board
3 印 *yìn* to print
4 成百上千 *chéngbǎi shàngqiān* hundreds or thousands
5 印刷 *yìnshuā* to print
6 唐代 *Tángdài* Tang dynasty (618–907)
7 金刚经 *Jīngāng Jīng* the Diamond Sutra (Buddhist text)
8 相当于 *xiāngdāngyú* to be equivalent to
9 敦煌 *Dūnhuáng* Dunhuang (oasis town in Gansu)
10 山洞 *shāndòng* cave
11 伦敦 *Lúndūn* London
12 大英图书馆 *Dàyīng Túshūguǎn* the British Library
13 雕版印刷 *diāobǎn yìnshuā* woodblock printing
14 北宋 *Běi Sòng* Northern Song dynasty (960–1127)
15 毕昇 *Bì Shēng* Bi Sheng, 11th-century inventor of movable type
16 工匠 *gōngjiàng* craftsman
17 黏土 *niántǔ* clay
18 硬 *yìng* hard
19 活字 *huózì* movable type

The same block could be printed again and again, so hundreds or thousands of copies of a book could be made. 5

The oldest surviving printed book with a clear date is a Tang-dynasty copy of the Diamond Sutra; the date on it corresponds to the year 868. It was found in a cave at Dunhuang and is kept today in the British Library in London. Woodblock printing had a drawback, though: every new book needed a whole new set of carved blocks, which cost a great deal of time and effort. 6

In the Northern Song, an ordinary craftsman named Bi Sheng came up with a new idea. He made many small blocks of clay, carved one character on each, and then hardened them in a fire: this was movable type. 7

印[1]书的时候，把需要的字一个一个排在铁板上，印完以后，还可以把字拆[2]下来，下次再用。毕昇[3]的故事被宋代[4]的学者沈括[5][6]记在了他的《梦溪笔谈[7]》里，所以我们今天才能知道。

但是，汉字的数量非常多，一套活字[8]要准备几千个甚至几万个字，所以在中国，雕版印刷[9]一直用得比活字更多。后来，人们又用木头和金属[10]做活字，朝鲜半岛[11]也很早就用上了金属活字。大约四百年以后，德国人古腾堡[12][13]在欧洲[14]做出了金属活字印刷机[15]。他是不是受到了中国的影响，历史学家到今天还在讨论。不管怎么样，印刷术[16]让书变得便宜，让更多的普通人有机会读书。

1 印 *yìn* to print
2 拆 *chāi* to take apart
3 毕昇 *Bì Shēng* Bi Sheng, 11th-century inventor of movable type
4 宋代 *Sòngdài* Song dynasty
5 学者 *xuézhě* scholar
6 沈括 *Shěn Kuò* Shen Kuo (1031–1095), Song scholar
7 梦溪笔谈 *Mèngxī Bǐtán* Dream Pool Essays, Shen Kuo's notebook
8 活字 *huózì* movable type
9 雕版印刷 *diāobǎn yìnshuā* woodblock printing
10 金属 *jīnshǔ* metal
11 朝鲜半岛 *Cháoxiǎn Bàndǎo* Korean peninsula
12 德国 *Déguó* Germany
13 古腾堡 *Gǔténgbǎo* Johannes Gutenberg (c. 1400–1468)
14 欧洲 *Ōuzhōu* Europe
15 印刷机 *yìnshuājī* printing press
16 印刷术 *yìnshuāshù* printing

To print, the characters needed were set one by one on an iron plate; after print- 7
ing, the type could be taken apart and used again next time. Bi Sheng's story was written down by the Song scholar Shen Kuo in his Dream Pool Essays; that is the only reason we know about it today.

But Chinese has a huge number of characters: a set of type needed thou- 8
sands or even tens of thousands of pieces, so in China woodblock printing always remained more common than movable type. Later, type was also made of wood and of metal, and the Korean peninsula was an early user of metal type. About four hundred years later, the German Gutenberg built a metal movable-type printing press in Europe. Whether he was influenced by China is something historians still discuss today. Either way, printing made books cheap and gave many more ordinary people the chance to read.

指南针[1]

很早以前，中国人就发现了一种奇怪的石头，它能吸住铁，这就是磁石[2]。根据古书的记载，两千多年前已经有了一种叫“司南[3]”的东西。它是用磁石做成的勺子[4]，放在一个光滑[5]的盘子上，转动以后，勺子的把儿最后会指向南方。不过，司南到底是什么样子，怎么用，学者[6]们的看法并不完全一样。

一开始，人们主要用这种工具来看风水[7]，比如决定房子和坟墓[8]的方向。到了宋代[9]，人们发现，用磁石摩擦[10]一根铁针，铁针也会有磁性[11]，而且比勺子灵敏[12]得多。

1 指南针 *zhǐnánzhēn* compass
2 磁石 *císhí* lodestone, magnet
3 司南 *sīnán* sinan, ancient “south-pointer”
4 勺子 *sháozi* spoon
5 光滑 *guānghuá* smooth
6 学者 *xuézhě* scholar
7 风水 *fēngshuǐ* fengshui
8 坟墓 *fénmù* tomb, grave
9 宋代 *Sòngdài* Song dynasty
10 摩擦 *mócā* to rub
11 磁性 *cíxìng* magnetism
12 灵敏 *língmǐn* sensitive

The Compass

Long ago, the Chinese discovered a strange kind of stone that could hold on 9
to iron: the lodestone. According to ancient books, more than two thousand years ago there was already a device called the sinan, the "south-pointer". It was a spoon made of lodestone, placed on a smooth plate; once spun, the spoon would come to rest with its handle pointing south. Scholars do not fully agree, however, on what the sinan really looked like or how it was used.

At first, such tools were used mainly for fengshui, for example to decide 10
which way a house or a tomb should face. By the Song dynasty, people had found that rubbing an iron needle with lodestone made the needle magnetic too, and far more sensitive than a spoon.

10 沈括[1]在书里也写到了这种磁针[2]，还记录了一个重要的发现：针指的方向并不是正南，而是稍微[3]偏[4]一点儿。人们把针放在水上漂[5]着，或者用丝线[6]挂起来，就做成了更好用的指南针[7]。

11 指南针最重要的用处[8]是在海上。在大海上，阴天的时候看不到太阳和星星，船很容易迷路[9]。大约在北宋[10]末年，中国的船员[11]已经开始用指南针来确定方向了。后来，阿拉伯[12]人和欧洲[13]人也用上了指南针。没有指南针，后来欧洲人的远洋[14]航行[15]就很难实现，世界地图也可能会是另[16]一个样子。

1 沈括 *Shěn Kuò* Shen Kuo (1031–1095), Song scholar
2 磁针 *cízhēn* magnetic needle
3 稍微 *shāowēi* slightly
4 偏 *piān* to lean to one side
5 漂 *piāo* to float
6 丝线 *sīxiàn* silk thread
7 指南针 *zhǐnánzhēn* compass
8 用处 *yòngchu* use
9 迷路 *mílù* to get lost
10 北宋 *Běi Sòng* Northern Song dynasty (960–1127)
11 船员 *chuányuán* sailor, crew
12 阿拉伯 *Ālābó* Arab, Arabia
13 欧洲 *Ōuzhōu* Europe
14 远洋 *yuǎnyáng* ocean-going
15 航行 *hángxíng* voyage, navigation
16 另 *lìng* another

Shen Kuo also wrote about such needles in his book, and recorded an important observation: the needle does not point due south, but slightly to one side. By floating the needle on water or hanging it from a silk thread, people made a much handier compass. 10

The compass's most important use was at sea. Out on the open sea, when the sky is overcast and neither the sun nor the stars can be seen, a ship easily loses its way. By around the end of the Northern Song, Chinese sailors had already begun to use the compass to find their direction. Later, Arabs and Europeans began to use the compass as well. Without the compass, the great ocean voyages the Europeans later made would hardly have been possible, and the map of the world might look quite different. 11

12 在中国，明代[1]的郑和[2]带着很大的船队七次下西洋[3]，最远到了非洲[4]的东海岸[5]。他们的航海图[6]上，写着每一段路应该按照指南针[7]的哪个方向走。这种记录航线[8]的方法，叫做“针路[9]”。

火药[10]

13 火药的发明其实是一个“意外”。古代有一些道士[11]想炼[12]出一种吃了以后能长生不老[13]的药。他们把各种东西放在炉子[14]里加热[15]，其中就有硝石[16]、硫磺[17]和木炭[18]。

1 明代 *Míngdài* Ming dynasty (1368–1644)
2 郑和 *Zhèng Hé* Zheng He (1371–1433), Ming admiral
3 下西洋 *xià Xīyáng* to sail the “Western Oceans”
4 非洲 *Fēizhōu* Africa
5 海岸 *hǎi'àn* coast
6 航海图 *hánghǎitú* nautical chart
7 指南针 *zhǐnánzhēn* compass
8 航线 *hángxiàn* sea route
9 针路 *zhēnlù* “needle route”, compass bearing
10 火药 *huǒyào* gunpowder
11 道士 *dàoshì* Daoist priest
12 炼 *liàn* to refine by heating
13 长生不老 *chángshēng bùlǎo* to live forever
14 炉子 *lúzi* furnace, stove
15 加热 *jiārè* to heat
16 硝石 *xiāoshí* saltpetre
17 硫磺 *liúhuáng* sulphur
18 木炭 *mùtàn* charcoal

In China, the Ming admiral Zheng He led a huge fleet on seven voyages to 12
the "Western Oceans", reaching as far as the east coast of Africa. Their sea charts noted, for each stretch of the way, which compass direction to follow. This way of recording a route was called a "needle route".

Gunpowder

The invention of gunpowder was really an "accident". In ancient times some 13
Daoists tried to produce an elixir that would let whoever took it live forever. They heated all kinds of substances in their furnaces, among them saltpetre, sulphur and charcoal.

13 他们发现，这三种东西放在一起，一点火就会突然燃烧，甚至爆炸[1]，有时候还会烧坏房子。大概在唐代[2]，已经有书提醒人们要小心这种危险的混合物[3]。

14 长生不老[4]的药没有炼[5]成，人们却很快发现了火药[6]的另[7]一个用处[8]：打仗[9]。宋代[10]的军队[11]用火药做成了火箭[12]、火球[13]和炸弹[14]。北宋[15]时期[16]的一本军事[17]书里，已经详细[18]地记录了几种火药的配方。后来，人们又做出了用竹筒[19]或者金属管子[20]发射[21]的武器[22]，这就是最早的枪[23]和炮[24]。

15 当然，火药也给人们带来了快乐。

1 爆炸 *bàozhà* to explode
2 唐代 *Tángdài* Tang dynasty (618–907)
3 混合物 *hùnhéwù* mixture
4 长生不老 *chángshēng bùlǎo* to live forever
5 炼 *liàn* to refine by heating
6 火药 *huǒyào* gunpowder
7 另 *lìng* another
8 用处 *yòngchu* use
9 打仗 *dǎzhàng* to fight a war
10 宋代 *Sòngdài* Song dynasty
11 军队 *jūnduì* army
12 火箭 *huǒjiàn* fire arrow (today: rocket)
13 火球 *huǒqiú* fire ball, incendiary ball
14 炸弹 *zhàdàn* bomb
15 北宋 *Běi Sòng* Northern Song dynasty (960–1127)
16 时期 *shíqī* period
17 军事 *jūnshì* military
18 详细 *xiángxì* detailed
19 竹筒 *zhútǒng* bamboo tube
20 金属 *jīnshǔ* metal
21 管子 *guǎnzi* tube, pipe
22 发射 *fāshè* to fire, to launch
23 枪 *qiāng* gun
24 炮 *pào* cannon

They found that when these three were mixed, a single spark made them suddenly burst into flame or even explode, sometimes burning down the building as well. By about the Tang dynasty, books were already warning people to be careful with this dangerous mixture. 13

The elixir of immortality was never achieved, but people soon found another use for gunpowder: war. Song armies used gunpowder to make fire arrows, fire balls and bombs. A military book from the Northern Song already recorded several gunpowder recipes in detail. Later, people made weapons that fired from bamboo tubes or metal barrels: these were the earliest guns and cannons. 14

Gunpowder brought people joy as well, of course. 15

15 过年的时候放的鞭炮[1]和烟花[2]，都是用火药[3]做的。大约在十三世纪，火药通过蒙古[4]人的战争传到了中东[5]和欧洲[6]。有了枪[7]炮[8]以后，以前的城墙[9]和铁甲[10]越来越没用，战争的样子也完全改变了。

不只是四个

16 有意思的是，“四大发明[11]”这个说法[12]并不是古代就有的。四百多年前，英国学者培根[13][14]写过，印刷术[15]、火药和指南针[16]这三种发明改变了整个世界，可是他并不知道它们都来自中国。后来，研究中国历史的西方学者把造纸术[17]也加了进去，到了二十世纪，“四大发明”才成为一个流行的说法。

1 鞭炮 *biānpào* firecrackers
2 烟花 *yānhuā* fireworks
3 火药 *huǒyào* gunpowder
4 蒙古 *Měnggǔ* Mongol, Mongolia
5 中东 *Zhōngdōng* Middle East
6 欧洲 *Ōuzhōu* Europe
7 枪 *qiāng* gun
8 炮 *pào* cannon
9 城墙 *chéngqiáng* city wall
10 铁甲 *tiějiǎ* iron armour
11 四大发明 *sì dà fāmíng* Four Great Inventions
12 说法 *shuōfǎ* expression, phrase
13 学者 *xuézhě* scholar
14 培根 *Péigēn* Francis Bacon (1561–1626)
15 印刷术 *yìnshuāshù* printing
16 指南针 *zhǐnánzhēn* compass
17 造纸术 *zàozhǐshù* papermaking

The firecrackers and fireworks set off at New Year are all made with gunpowder. Around the thirteenth century, gunpowder spread to the Middle East and Europe through the wars of the Mongols. Once there were guns and cannons, the old city walls and iron armour became less and less useful, and the face of war changed completely. 15

Not Just Four

Interestingly, the phrase "Four Great Inventions" does not go back to ancient times. Some four hundred years ago the English thinker Francis Bacon wrote that three inventions – printing, gunpowder and the compass – had changed the whole world; but he did not know that all three came from China. Later, Western scholars of Chinese history added papermaking to the list, and only in the twentieth century did "the Four Great Inventions" become a popular phrase. 16

17 其实，中国古代的发明远远[1]不止[2]这四个。丝绸[3]、瓷器[4]、算盘[5]、纸币[6]，还有张衡[7]的地动仪[8]，都是中国人的创造。而且，一项技术能改变世界，不只是因为它被发明了，还因为很多不同国家的人学习它、改进它，最后把它带到了世界各地。四大发明[9]的故事，也是不同文化互相学习的故事。

1 远远 *yuǎnyuǎn* far (more)
2 不止 *bùzhǐ* more than, not limited to
3 丝绸 *sīchóu* silk
4 瓷器 *cíqì* porcelain
5 算盘 *suànpán* abacus
6 纸币 *zhǐbì* paper money
7 张衡 *Zhāng Héng* Zhang Heng (78–139), Han astronomer
8 地动仪 *dìdòngyí* seismoscope
9 四大发明 *sì dà fāmíng* Four Great Inventions

In fact, ancient China invented far more than these four things. Silk, porcelain, the abacus, paper money and Zhang Heng's seismoscope were all Chinese creations. And a technology changes the world not only because it was invented, but because people in many different countries learned it, improved it and finally carried it to every part of the world. The story of the Four Great Inventions is also a story of different cultures learning from one another. 17